

1 Preliminaries

We describe in detail the numerical algorithm used in Example 4 from Section 7 of our paper. For this, we first recall some relevant main results from the paper. We refer to the paper for proofs of the following results.

We assume throughout that the aggregation function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ has the following decomposition:

$$g(\mathbf{x}) = \tilde{g}(\mathbf{x}^{(1)}) + \boldsymbol{\beta}^T \mathbf{x}^{(2)}, \quad (1)$$

where $\mathbf{x} = (\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ with $\mathbf{x}^{(1)} \in \mathbb{R}^m$, $\mathbf{x}^{(2)} \in \mathbb{R}^{n-m}$, for some $m \leq n$, $\boldsymbol{\beta} \in \mathbb{R}^{n-m}$, and $\tilde{g} : \mathbb{R}^m \rightarrow \mathbb{R}$ is Lipschitz continuous.

The next result characterizes the solution to the following problem:

$$\max_{G \in \mathcal{M}_p(\mathbb{R})} \tilde{I}_h(G), \quad \text{subject to} \quad \mathcal{B}_\phi(G^{-1}, F^{-1}) \leq \varepsilon, \quad (2)$$

where for the univariate BW divergence, we write $\mathcal{B}_\phi(F_1^{-1}, F_2^{-1}) = \mathcal{B}_\phi[F_1, F_2]$.

Theorem 1 *Let I_h be a signed Choquet integral with absolutely continuous distortion function h and distortion weight function γ . Suppose ϕ is strictly convex and differentiable with $\lim_{x \rightarrow \infty} \phi'(x) = \infty$ and $\lim_{x \rightarrow -\infty} \phi'(x) = -\infty$. Assume at least one of the following holds:*

- i) γ is non-decreasing*
- ii) γ is non-negative and $\int_0^1 \phi'(F^{-1}(u))^2 + \gamma(u)^2 du < \infty$.*

Then, the worst-case quantile function of (2) is uniquely given by

$$G_{\lambda^*}^{-1}(u) := (\phi')^{-1} \left(\left(\phi'(F^{-1}(u)) + \frac{1}{\lambda^*} \gamma(u) \right)^\uparrow \right), \quad (3)$$

where $\lambda^ > 0$ is the smallest solution to $\mathcal{B}_\phi(G_{\lambda^*}^{-1}, F^{-1}) = \varepsilon$ in case i) and the unique solution to $\mathcal{B}_\phi(G_{\lambda^*}^{-1}, F^{-1}) = \varepsilon$ in case ii).*

We also note that in case i), the function $\phi'(F^{-1}(u)) + \frac{1}{\lambda^*} \gamma(u)$ is non-decreasing and equal to its isotonic projection.

The following two results will also be relevant to our main algorithm.

Proposition 1 Suppose that I_h satisfies the conditions of Theorem 1. Further, assume that the aggregation function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is K -Lipschitz with decomposition (1) and the distortion weight function γ satisfies $0 < \|\gamma\|_2 < \infty$. Let $\mathbf{X} \in \mathcal{L}_n^2$ such that $g(\mathbf{X}) \in \mathcal{L}^2$ and $\varepsilon > 0$.

i) If γ non-decreasing, then

$$I_h(g(\mathbf{X})) + \|\beta\|_2 \|\gamma\|_2 \varepsilon \leq \sup_{\mathbf{Y} \in \mathcal{U}_\varepsilon^n(\mathbf{X})} I_h(g(\mathbf{Y})) \leq I_h(g(\mathbf{X})) + K \|\gamma\|_2 \varepsilon.$$

ii) If γ is non-negative, then

$$\int_0^1 \gamma(u) \left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\lambda} \gamma(u) \right)^\uparrow \leq \sup_{\mathbf{Y} \in \mathcal{U}_\varepsilon^n(\mathbf{X})} I_h(g(\mathbf{Y})) \leq \int_0^1 \gamma(u) \left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\bar{\lambda}} \gamma(u) \right)^\uparrow du, \quad (4)$$

where λ is the unique positive solution to

$$\int_0^1 \left(F_{g(\mathbf{X})}^{-1}(u) - \left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\lambda} \gamma(u) \right)^\uparrow \right)^2 du = \|\beta\|_2^2 \varepsilon^2,$$

and $\bar{\lambda}$ is the unique positive solution to

$$\int_0^1 \left(F_{g(\mathbf{X})}^{-1}(u) - \left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\bar{\lambda}} \gamma(u) \right)^\uparrow \right)^2 du = K^2 \varepsilon^2. \quad (5)$$

Proposition 2 (Ordering property) Let $l_1, l_2 \in \mathcal{L}^2((0, 1))$. If $l_2(s) \leq l_1(s)$ for all $s \in (0, 1)$, then $l_2^\uparrow(s) \leq l_1^\uparrow(s)$. This result also holds when the inequalities are replaced with strict inequalities.

2 Description of Algorithm

We propose Algorithm 1 to compute the optimal Lagrange parameter $\bar{\lambda}$.

Algorithm 1 Optimal Lagrange multiplier

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1: Initialize  $\lambda^{(0)} = \frac{\|\gamma\|_2}{2K\varepsilon}$ .
2: Input: Error tolerance  $c \geq 0$ , step size  $h_t > 0$ .
3:  $t \leftarrow 0$ 
4: Compute the isotonic projection of  $H_{\lambda^{(t)}}(u)$ .
5: Compute

$$I(\lambda^{(t)}) := \int_0^1 \left( F_{g(\mathbf{X})}^{-1}(u) - H_{\lambda^{(t)}}(u)^\uparrow \right)^2 du.$$

6: while  $I(\lambda^{(t)}) < K^2\varepsilon^2$  or  $I(\lambda^{(t)}) > K^2\varepsilon^2 + c$  do
7:   if  $I(\lambda^{(t)}) < K^2\varepsilon^2$  then
8:      $\lambda^{(t+1)} \leftarrow \lambda^{(t)} - h_t$ 
9:   else
10:     $\lambda^{(t+1)} \leftarrow \lambda^{(t)} + h_t$ 
11:   end if
12:   Compute the isotonic projection of  $H_{\lambda^{(t+1)}}(u)$ .
13:   Compute  $I(\lambda^{(t+1)})$ .
14:    $t \leftarrow t + 1$ 
15: end while
16: Return:  $\hat{\lambda} = \lambda^{(t)}$ .
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3 Remarks and Theoretical Justifications

We make the following remarks about our algorithm. First, the initialization is the Lagrange multiplier when γ is non-decreasing. Thus, if $H_{\lambda^{(0)}}(u) := F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\lambda^{(0)}}\gamma(u)$ is non-decreasing, then it is equal to $G_{\bar{\lambda}}^{-1}$, in which case, the algorithm converges immediately to the correct value (i.e., $\hat{\lambda} = \lambda^{(0)} = \bar{\lambda}$). Second, the step size h_t should be a non-constant and non-increasing sequence. If h_t is constant, then the algorithm might oscillate between two values and never converge. Finally, if the algorithm converges to $\hat{\lambda}$, then $\tilde{I}_h(G_{\hat{\lambda}}^{-1})$ is an upper bound for $\tilde{I}_h(G_{\bar{\lambda}}^{-1})$ (and therefore an upper bound for $\sup_{\mathbf{Y} \in \mathcal{U}_\varepsilon^n(\mathbf{X})} I_h(g(\mathbf{Y}))$). We state this formally in the following proposition.

Proposition 3 *Suppose that the conditions of Proposition 1 ii) hold. If Algorithm 1 converges to $\hat{\lambda}$, then $\tilde{I}_h(G_{\hat{\lambda}}^{-1}) \leq \tilde{I}_h(G_{\bar{\lambda}}^{-1})$. Furthermore, if $c = 0$, then the inequality becomes an equality.*

Proof: By definition of G_λ^{-1} ,

$$\tilde{I}_h(G_{\hat{\lambda}}^{-1}) - \tilde{I}_h(G_{\bar{\lambda}}^{-1}) = \int_0^1 \gamma(u) \left[\left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\hat{\lambda}} \gamma(u) \right)^\dagger - \left(F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\bar{\lambda}} \gamma(u) \right)^\dagger \right] du.$$

It suffices to show that the second term in the integrand is non-negative.

The convergence criteria of Algorithm 1 implies that $I(\bar{\lambda}) = K^2 \varepsilon^2 \leq I(\hat{\lambda}) \leq K^2 \varepsilon^2 + c = I(\bar{\lambda}) + c$. Recall from the proof of Theorem 1 *ii*) that I is a strictly decreasing function of λ . Hence, it must be the case that $\hat{\lambda} \leq \bar{\lambda}$. Thus, $F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\hat{\lambda}} \gamma(u) \geq F_{g(\mathbf{X})}^{-1}(u) + \frac{1}{2\bar{\lambda}} \gamma(u)$ and the desired inequality for the isotonic projections follows from Proposition 2. Finally, for the case of $c = 0$, we have $\hat{\lambda} = \bar{\lambda}$, so the inequality becomes an equality. ■

The key step in the algorithm is computing the integral $I(\lambda^{(t)})$ via numeric integration, which requires evaluating the isotonic projection of $H_{\lambda^{(t)}}$ on a finite partition of $(0, 1)$. Since we use a finite partition, the isotonic projection reduces to the isotonic regression, which can be calculate using active set methods (de Leeuw et al. (2009)). We perform our computations in R using the function *activeSet* from the package *isotone*.

Note that Algorithm 1 can also be used to compute the lower bound in (4) with two minor adjustments. First, we replace K with $\|\beta\|_2$. Second, we change the convergence criteria to $\|\beta\|_2^2 \varepsilon^2 - c \leq I(\lambda^{(t)}) \leq \|\beta\|_2^2 \varepsilon^2$ for some error tolerance $c \geq 0$. This convergence criteria guarantees that if the algorithm converges to $\hat{\lambda}$, then $\hat{\lambda} \geq \underline{\lambda}$, and $\tilde{I}_h(G_{\hat{\lambda}}^{-1})$ is a lower bound by the arguments in Proposition 3.

References

- de Leeuw, J., Hornik, K., and Mair, P. (2009). Isotone optimization in R: Pool-adjacent-violators algorithm (PAVA) and active set methods. *Journal of Statistical Software*, 32(5):1–24.